CHAPTER 5

SECURITY

FROM THE RECALIBRATION of military strategy to the reconstitution of diplomacy, AI will become a key determinant of order in the world. Immune to fear and favor, AI introduces a new possibility of objectivity in strategic decision-making. But that objectivity, harnessed by both the warfighter and the peacemaker, should preserve the subjectivity of human wisdom that is essential for the responsible exercise of force. More a revelation of our existing condition than a further discovery of the unknown, AI in war will illuminate the best and worst expressions of our humanity. Even before the implementation of serious breakthroughs, an understanding will emerge of the extent to which AI can be, at once, the means of a conflict and the architect of its termination.

Humanity's long-standing struggle to constitute itself in ever-more complex arrangements, so that no state shall be in a position to have absolute mastery over others, has achieved the status of a continuous, uninterrupted law of nature. In a world where the major actors are still human—even if equipped with AI to inform, consult, and advise them—we should still enjoy a degree of stability based on shared norms of conduct among the actors involved, subject to the tunings and adjustments of time.

But if AI emerges as a practically independent political, diplomatic, and military set of entities, that would force the exchange of the age-old balance of power for a new, uncharted disequilibrium. The international concert of nation-states—a tenuous

and internally shifting equilibrium achieved in the last few centuries—has held in part because of the inherent equality of the players. A world of severe asymmetry—for instance, if some states adopted AI into political leadership more readily than others—would be far less predictable. In cases where some humans might face off militarily or diplomatically against a highly AI-enabled state, or against AI itself, humans could seem defenseless to survive, much less compete. Such an intermediate order could witness an internal implosion of societies and an uncontrollable explosion of external conflicts.

Beyond a minimum of security, humans have long entered into combat in pursuit of triumph or in defense of honor. But machines—for now—lack any real conception of either triumph or honor. They might prosecute a kind of war unique in human experience. But what choices would AIs make? Might they never go to war, choosing instead (for instance) immediate, carefully divided transfers of territory based on complex calculations of relative strategic advantage? Or might they—prizing an outcome and deprioritizing individual lives—spiral into wars of human attrition? In one scenario, our species could emerge so transformed as to avoid entirely the brutality of human conduct. In another, we would become so subjugated as to return us to a barbaric past.

Is there no other way? Thus far, neither the patient overtures of diplomacy nor the shocking horrors of war have ever succeeded in permanently encoding in humans—let alone another species—an aversion to destruction. Is it nevertheless possible, thanks to the age of AI, that conditions will finally materialize for a perpetual peace to emerge at last?

ESPIONAGE AND SABOTAGE

Even as nations search for ways to keep the technology safe, they are simultaneously fixated on how to "win the AI race."[1] In part, that response is understandable. Culture, history, communication, and perception have conspired to create among today's primary states a

diplomatic situation that fosters insecurity and suspicion on all sides. AI, as a dominant feature of the global future, forms a low flashpoint in an already volatile combination in which each side believes that an incremental tactical advantage could be decisive for its long-term benefit.

If, following from the instinct for self-preservation, each human society wishes to maximize its unilateral position, then the conditions would be set for a psychological contest between rival military forces and intelligence agencies, the likes of which humanity has never faced before. Today, in the years, months, weeks, and days leading up to the arrival of the first superintelligence, a security dilemma of existential nature awaits. The logical first wish for any human actor coming into possession of such a capability might be to attempt a guarantee of singular continued existence. Any such actor might also reasonably assume by default that its rival, under the same uncertainties and facing the same stakes, would be pondering a similar move.

Even if a dominant nation were to stop short of war, a superintelligent AI could subvert, undermine, and block a competing program. For instance, AI promises both to strengthen conventional viruses with unprecedented potency and to disguise them with similar thoroughness. Like the computer worm Stuxnet—the cyberweapon said to have ruined a fifth of Tehran's uranium centrifuges before detection—an AI agent could sabotage a rival's progress in ways that obfuscate its presence, thereby leading enemy scientists down futile pathways of inquiry.[2]

With its unique capacity for manipulation of weaknesses in human psychology *en masse*, an AI could also hijack a rival nation's media, producing a deluge of synthetic disinformation so alarming as to inspire mass opposition against further progress in that country's AI capacities. Or, to another nation's chief AI scientist, it might target and personalize communications to render his or her perception of personal grief so great as to degrade the capacity for effective leadership.

Assessing the state of the competition will be even more

challenging. Already the largest AI models are being trained on secure networks disconnected from the rest of the internet. Some executives believe that AI development will itself sooner or later migrate to impenetrable bunkers whose supercomputers are powered with nuclear reactors.[3] Data centers are even now being built on the bottom of the ocean floor.[4] Soon they could be sequestered in orbits around Earth. Corporations or nations might increasingly "go dark," ceasing to publish AI research so as not only to avoid enabling malicious actors (as publicly stipulated) but also to obscure their own pace of construction (as privately intended). To distort the true picture of their progress, others might even try deliberately publishing misleading research, with AI assisting in the creation of convincing fabrications.

There is a precedent for such scientific subterfuge: In 1942, the Soviet physicist Georgy Flyorov correctly inferred that the U.S. was building a nuclear bomb after he noticed that the Americans and the British had suddenly stopped publishing scientific papers on atomic fission.[5] Today, however, such a contest would be made all the more unpredictable given the complexity and ambiguity of measuring progress toward something so abstract as intelligence. Although some see "advantage" as commensurate with the size of the AI models in their possession, a larger model is not necessarily superior across all contexts and may not always prevail over smaller, inferior models deployed at scale. Smaller and more specialized AI machines might operate like a swarm of drones against an aircraft carrier—unable to destroy it, but sufficient to neutralize much of its capacity.

To some actors, an overall advantage would be signaled by the achievement of a given capability. The problem with this line of thinking is that *AI* refers merely to a process of machine learning that is embedded not just in a single technology but in a broad spectrum of technologies. Capability in any one area may thus be driven by factors entirely different from capability in another. In these senses, any "advantage" as ordinarily calculated may be illusory.

Moreover, as demonstrated by the exponential and unforeseen

explosion of AI capability in recent years, the trajectory of progress is neither linear nor predictable. Looking ahead, experts continue to disagree about the development of superintelligence: Is it merely a matter of scaling and the implementation of existing learning infrastructures, or will superintelligence require additional creative and scientific innovations?[6] Conceivably, the transition from narrow intelligence to general intelligence and then to superintelligence could pass without obvious signs of evolution—particularly if humans do not develop a uniform conception of what to look for. Even if one actor could be said to "lead" another by an approximate number of years or months, a sudden technical or theoretical breakthrough in a key area at a critical moment could invert the positions of all players.

In such a world, where no leaders could trust their most solid intelligence, their most primal instincts, or even the basis of reality itself, no real contender for superintelligence could be blamed for acting from a position of maximum paranoia and suspicion. Leaders are no doubt already making decisions under the assumption that their endeavors are under surveillance or harbor distortions created by malign influence. Defaulting to worst-case scenarios, the strategic calculus of any actor at the frontier would be to prioritize speed and secrecy over safety. With human perception unable to detect or defend against AI-enabled attempts at suppression, human leaders could be gripped by the fear that there is no such thing as second place. Under pressure, they might prematurely accelerate the deployment of AI as deterrence against external disruption.

Today, we are still behind an involuntary veil of ignorance. The ultimate winners (if winning can be defined) of the competition for AI dominance are as yet unknown. Every enterprise in pursuit of the front rank is a potential rival. This uncertainty will produce instability.

Traditionally, after the emergence of a new power, a bloody contention has been necessary before rivals can settle into a new status quo that is at least minimally acceptable to all parties. However, in a world of nuclear weapons complicated by such a rapid reconfiguration by AI, there may be no opportunity to validate a new consensus by the established principles and instruments of warfare.

In the event that the identity of a winner does crystallize, mere competition could devolve into conflicts driven by desperation and fear. In such an instance, certitude could be even more destabilizing than precaution. Put together, perfect deterrence with unlimited speed and maximum accuracy translates to total victory. Historical pursuits of a monopoly on force have typically caused an attendant expansion in the assembly of dark designs by others. When such capacities are considered in the context of a precariously balanced world, some states may deem the advent of AI threatening enough to demand a nuclear response. Having obviated conventional wars, will AI then drag us backward into nuclear ones?

In previous arms races, evolutionary instinct mixed with competition among inventors has enabled the creation and operation of instruments designed to preserve homeland societies. Here, too, defensive AI systems could neutralize attacks by adversaries, for instance by updating currently vulnerable software and other systems, or, if tasked with the surveillance of competing programs, by serving as an early warning system. However, in this case, new threats—an AI-enabled bioweapon, a sudden nuclear proliferation, or even a misaligned AI itself—might emerge so quickly and silently, and wreak such devastation, as to preempt any response.

Though we may not yet have entered such a moment, preparations should be made in advance to manage the existential competition of the AI age and its attendant risks. A determined or desperate actor with secondary but sizable strength will be vigilant in the watch for superior AIs. If that secondary actor perceives—even mistakenly—that the dominant other is within grasp of a totalizing capability, it may launch preemptive electronic or physical strikes, setting off a cycle of unimaginable escalation and retaliation and the spiraling possibility of mutual destruction.

A glimmer of hope—at least for today—lies in our uncertainty. Ambiguity can be a fertile ground in which to cultivate dialogue. Unsure of how any one nation or group might be advantaged or disadvantaged by AI decisions made today, global leaders have a window of opportunity to engage in discussions on the basis of humanity's collective survival.

MANAGING EMERGENCE

If the future is a competition to reach a single, perfect, unquestionably dominant intelligence, then it seems likely that humanity will either lose control of an existential race among multiple actors or suffer the exercise of supreme hegemony by a victor unharnessed by traditional checks and balances. The closer the margin by which the race is won, the higher the probability of human miscalculation.

Unipolarity may be one pathway that could minimize the risk of extinction. Were the existing front-runner able to extend its lead to a threshold where no other entity believed it possible to close the gap, there might be greater certainty of ensuring some level of stability. At least for a while, the basis of world order—the ceaseless pursuit of transitory and fragile equilibria among balanced forces—might no longer be desirable.

Alternatively, it might be possible—though contingent on the suppression of natural human instincts—to broker an agreement among competing entities to assure an agreed-upon period of cooperation or even to merge into a single enterprise the multiple competing efforts at the head of the line. But achieving unification across geopolitical and commercial hostilities would require an extraordinary amount of courage and foresight. Any actor at or near the front would expect to be included in the company of victory. Within that prime category, indecisive or still-lagging actors would naturally achieve the greatest benefit from entering into the arrangement, thereby becoming well placed to gain parity and possibly to pull ahead. By contrast, other leading actors might find the sacrifice of their pole position intolerable, and their trust in altruism might prove too brittle to withstand the temptations of outright defection.

A similar option would be a negotiation to distribute and delegate power. When confident that victory is imminent, the best-positioned actor might attempt to persuade its rivals to surrender by guaranteeing their privileged access to superintelligent AI. But never

has trust in the human commitment to one's fellow man, much less to one's former adversary, endured forever. If humans were to manage, against all odds, to forge such an agreement, it is far from clear how its implementation would be governed.

Indeed, simply collecting the sharpest peaks on the "island" of knowledge might exacerbate the dynamics that their consolidation was designed to prevent in the first place. Human history has yet to produce any record of such an attempt, much less a success. Moreover, such an arrangement would require a dramatic reorientation of diplomatic strategy. Historically, the very basis of world order has been maintained by the ceaseless pursuit of a delicate equilibrium of balanced forces. Here, by contrast, nation-states would be pursuing a hegemonic stasis largely unfamiliar to human practice. The dominant bloc of states likely would be the original creators of AI systems and the suppliers of their most critical components, as well as the producers of the talent to develop and refine them. The nondominant majority might be relegated to the status of tributaries, supplying data and other goods and repaid with some level of access to AI's discoveries, governance systems, and defenses.

In any case, this scenario is neither our preference nor our prediction. Rather, we believe there will be not just one supreme AI but rather multiple instantiations of superior intelligence in the world. In that case, a different set of possible futures emerges. Our strongest creations, acting as countervailing forces, could be better equipped than humans to exert and maintain an equilibrium in global affairs inspired (but not constrained) by human precedent. Nonhuman intelligence could thus manage its own emergence, at least in the realms of national security and geopolitics.

But could AI engineer a sustainable division of dominion? Possibly, yes. Diplomatic negotiations, underneath their layers of human emotion and psychology, are at their core a species of applied game theory, which is itself a branch of mathematics. Even if the practice of diplomacy originated as a form of art (albeit one exclusive to the realm of human conduct), it might increasingly morph into a science. As it does, it could transcend the rather mixed

track record of human diplomacy when it comes to identifying and pursuing opportunities for compromise. Already, early AI models in the West have shown great promise in their ability to apply strategy, at least in gameplay, while China has moved a step further, leveraging machine intelligence to discharge the duties of human diplomats.[7]

The human tradition of diplomacy began with the simple need to convey messages between societies safely and reliably. Gradually, envoys from afar would come to enjoy special treatment, while states violating the new norms would seldom escape sanction. When representatives of Xerxes demanded symbols of submission from the Greek city-states, the Persian diplomats were thrown into pits and down wells. Herodotus records that the gods would later punish Sparta for this deed.[8] About seventeen centuries later, when a senior Mongol diplomat was executed on the order of a Persian shah, Genghis Khan commanded his fearsome horde to destroy the entire Khwarezmid empire in retribution.[9] Still, keeping open the channels of communication, even—and especially—in times of war, became custom; over time, a basic consensus formed that it might be better to hear the message than to kill the messenger.

Were AIs to conduct diplomacy among themselves, they might be intentionally trained to accept the same custom, or they might incidentally develop a bias in favor of the additional information that these customs guaranteed. Surely there would also be aberrations. While perhaps no heads would roll, machines might find an equivalent way of pursuing their narrow interests. Here their total rationality, if humans could accept it, might increase the probability of at least a safer start.

The provision of machine solutions to human problems as fundamental as diplomacy and security will, however, lead naturally to further dependence on AI's abilities—a dependence difficult to break if and when decisive human intervention in international affairs becomes necessary. At least human statecraft, however imperfectly applied, enforced upon us the responsibility for our own choices. Choosing to depend on AI, by contrast, may reduce our

ability to trust the basic human judgment on which we could reliably subsist in our more predictable but primitive past.

Why take such a risk? For one thing, to avoid the dilemma of mutually destructive competition or a hegemony produced by emergent superintelligence, but also to protect ourselves from other threats looming on the horizon. With every passing year, with each new technological breakthrough, the minimum threshold for destroying all of human society has lowered.

Nordic mythology tells the story of Baldur, son of the god Odin and Queen Frigg, alarmed by prophecies of their son's impending gruesome death.[10] Queen Frigg, determined to shield him from whatever fate lay ahead, traveled tirelessly across the nine realms of Earth, casting powerful magic on every animal, element, plant, and plague so that none could be used as a weapon against him. But the trickster god Loki, disguised as an old woman, coaxed from the queen the information that her divine cloak of protection covered everything on Earth except mistletoe, the most harmless weed of all. At a feast held to celebrate Baldur's immunity, the gods, to showcase the queen's fine work, took turns launching at him weapons of every make. Loki, however, compelled his blind brother Hodr to shoot an arrow tipped with mistletoe, piercing Baldur's chest and killing him by means of the only object in Valhalla exempt from the queen's otherwise comprehensive guardianship.

The meaning of this ancient myth for our modern predicament is simple and chilling: As threats grow ever more obfuscated and sophisticated, humanity's defenses against them must be ever more perfect, since the slightest mistake or omission could spell defeat. And to achieve that level of perfection, we might well need the assistance of AI.

The exercise is thus to decide which risk is lower and therefore should come first: surviving the development of AI, surviving other parallel revolutions like the advent of synthetic biology, or surviving potential catastrophes such as radical climate change. Granted, smaller AIs may assist in the reckless invention of existential technologies; even if the creators of these new means were

themselves perfectly and properly regulated, others would likely be less careful and more harmful. But larger AIs could assist in defending against the same technologies, enabling actual perfection in tactical defensive decision-making.

For instance, if developed, individualized biodefense—whereby AI-enabled nanobots would be deployed in our bloodstream to remove anything that failed to match a recognized biosignature—would be a more agile corrective than our previous response to biological threat. Similarly, AI may generate new materials and new processes to reduce carbon emissions, lowering the risk of climate catastrophes.

No doubt, it is a risk for AI to assume early and sustained responsibility for the species and societies behind its own conception, but the traditional pathways, which require perfection in human performance, may be even riskier. Best, in our current view, would be to have AI working before and not after humanity has to confront the proliferation of new threats to survival.[11] The appropriate question under this assumption is this: How can humans accelerate only desirable pathways for AI, while delaying the undesirable?

AI is an unselectively destabilizing force; its emergence, if unmanaged, poses as great a risk to its creators as to its users. This is precisely what might compel initially unwilling rivals to consider otherwise implausible agreements. We believe that in diplomacy, defense, and perhaps elsewhere, some of the risks of AI can be managed successfully only by AI itself. Pandora's box has already been opened; even if it had not been, the benefits of AI still appear to outweigh the risks.

It is our contention, therefore, that our "vulnerable world" (to borrow a phrase from Nick Bostrom) may well require intervention by AI in order to survive some of the challenges and threats of AI's technological development.[12] The question remains: How must humans behave when facing a future that simultaneously demands and forbids our continued control?

A NEW PARADIGM OF WAR

For almost all of human history, war has been fought in a defined space in which one could know with reasonable certainty the capability and position of hostile enemy forces. The combination of these two attributes offered each side a sense of psychological security and common consensus, allowing for the informed restraint of lethality. Only when enlightened leaders were unified in their basic understanding of how a war *might* be fought could opposing forces control *whether* a war should be fought.

Speed and mobility have been among the most predictable factors underpinning the capability of any given piece of military equipment. An early illustration is the development of the cannon. For a millennium after their construction, the Theodosian Walls protected the great city of Constantinople from outside invaders. Then, in 1452, an artillery engineer from the kingdom of Hungary—at that point under Byzantine suzerainty—proposed to Emperor Constantine XI the construction of the Basilic, a giant cannon that, firing from behind the defensive walls, would pulverize outside intruders. But the complacent emperor, possessing neither the material means nor the foresight to recognize the technology's significance, dismissed the proposal.

Unfortunately for him, the Hungarian engineer turned out to be a political mercenary: Switching tactics (and sides), he updated his design to be more mobile—transportable by no fewer than 60 oxen and 400 men—and approached the emperor's rival, Turkish Ottoman sultan Mehmed II, who was preparing to besiege the impermeable fortress. Winning the young sultan's interest with his claim that this gun could "shatter the walls of Babylon itself," the entrepreneurial Hungarian helped the Turkish forces to breach the ancient walls in only 55 days.[13]

The contours of this fifteenth-century drama can be seen again and again throughout history. In the nineteenth century, speed and mobility transformed the fortunes first of France, as Napoleon's Grande Armée overwhelmed Europe, and then of Prussia, under the

direction of Helmuth von Moltke (the Elder) and Albrecht von Roon, who capitalized on the newly developed railways and embraced distributed control to enable faster and more flexible maneuvering. Similarly, blitzkrieg—an evolution of the same German military principles—would be used against the Allies in World War II to great and terrible effect.

"Lightning war" has taken on new meaning—and ubiquity—in the era of digital warfare. Speed is instantaneous. Attackers need not sacrifice lethality to sustain mobility, as geography is no longer a constraint. Although that combination has largely favored the offense in digital attacks, an AI era could see the increase of the velocity of response and re-enable cyber defenses on par with cyber offenses.

In kinetic warfare, AI will provoke another leap forward. Drones, for instance, will be extremely quick and unimaginably mobile. Once AI is deployed not only to guide one drone but to man fleets of them, clouds of drones will form and fly in sync as a single cohesive collective, perfect in their synchronicity. Future drone swarms will dissolve and reconstitute themselves effortlessly in units of every size, much as elite special-operations forces are built from scalable detachments, each of which is capable of sovereign command.

In addition, AI will provide similarly speedy and flexible defenses. Drone fleets are impractical if not impossible to shoot down with conventional projectiles. But AI-enabled guns firing rounds of photons and electrons (instead of ammunition) could re-create the same lethal disabling capacities as a solar storm that can fry the circuitry of exposed satellites. Once more, speed and mobility will be beyond human capacity—and potentially equalized on defense and offense.

With speed and mobility no longer defining variables, the gap in capability between competing entities would now rely on precision, immediate impact, and strategic application.

AI-enabled weapons will be unprecedentedly exact. Limits to the knowledge of an antagonist's geography have long constrained the capabilities and intentions of any party to hostilities. But the alliance between science and war has come to ensure increasing accuracy in

our instruments, and AI can be expected to make another breakthrough—or many. AIs will thus shrink the gap between original intent and ultimate outcome, including in the application of lethal force. Whether land-based drone swarms, machine corps deployed in the sea, or possibly interstellar fleets, machines will possess highly precise capabilities of killing humans with little degree of uncertainty and with limitless impact. The bounds of the potential destruction will hinge only on the will, and the restraint, of human and machine.

That being so, the AI age of warfare will be reduced primarily to an assessment not of an adversary's capabilities but rather of its intentions and their strategic applications. In the nuclear age, we have already entered such a phase, in a sense—but its dynamics and significance will come into much sharper focus as AI proves its worth as a weapon of war. Thus, the key question is this: What will AI-enabled commanders want and need?

With such valuable technology at stake, humans are not likely to be considered the primary targets of AI-enabled war. AIs could in fact remove humans as a proxy in warfare entirely, making war less deadly but potentially no less decisive. Similarly, territory alone seems unlikely to provoke AI aggression—but data centers and other critical digital infrastructure certainly could. (Supercomputers are likely to be hidden and storage of intelligences distributed to secure a stronger chance of operational continuity and to defend against a mechanical "decapitation strike.")[14]

Surrender, then, will come not when the opponent's numbers are diminished and its armory empty but when the survivors' shield of silicon is rendered incapable of saving its technological assets, and finally its human deputies. War could evolve into a game of purely mechanical fatalities, the deciding factor being the psychological strength of the human (or AI) who must contest to risk, or forfeit to prevent, a breakthrough moment of total destruction.

Even the motives governing the new battlefield would be alien, to some extent. G. K. Chesterton tells us "the true soldier fights not because he hates what is in front of him, but because he loves what

is behind him."[15] An AI war is unlikely to involve love or hate, let alone a concept of soldierly bravery. On the other hand, it may still incorporate ego, identity, and loyalty—though the nature of those identities and loyalties may not be consistent with today's.

The calculation in warfare has always been relatively straightforward: Whichever side first finds intolerable the excruciations caused by its antagonist's might becomes subject to conquest and conversion, and only then, perhaps, to negotiation. The consciousness of one's own shortcomings reliably and naturally has produced restraint. Without such awareness, and with no sense for (and thus with total tolerance of) pain, one cannot but wonder what, if anything, would prompt restraint in an AI that has been introduced into warfare, and what would conclude the conflicts it wages. Would a chess-playing AI, if it had never been informed of the rules dictating the end of the game, have played to the last pawn?

GEOPOLITICAL RESTRUCTURING

In every age of humanity, almost as if in obedience to some natural law, there has emerged, as one of us once put it, a unit "with the power, the will, and the intellectual and moral impetus to shape the entire international system in accordance with its own values."[16] Subsequent to that entity's appearance, other units become linked in novel arrangements, building unpredictable dependencies in times of crisis and constantly threatening to undo geopolitical balances of power. In some cases, the resultant system upends existing authorities; in others, it entrenches them.

The most familiar arrangement of human civilizations is that of the Westphalian system as conventionally understood. The idea of the sovereign nation-state, however, is only a few centuries old, having emerged from treaties that are collectively known as the Peace of Westphalia in the mid-seventeenth century. It is not the preordained unit of social organization, and it may not be suited for

the age of AI. Indeed, as mass disinformation and automated discrimination trigger a loss of faith in that arrangement, AI may pose an inherent challenge to the power of national governments. Compounding the issue would likely be the psychological disorientation and possible retreat from reality detailed earlier in this book. Alternatively, AI may well reset the relative positions of competitors within today's system. If its powers are harnessed primarily by nation-states themselves, humanity could be forced toward a hegemonic stasis, or else toward a new equilibrium of AI-empowered nation-states. But it could also be the catalyst of an even more fundamental transition—a shift to an entirely new system, in which state governments would in turn be forced to abandon their central role in the global political infrastructure.

One possibility is that the companies that own and develop AI will accrue totalizing social, economic, military, and political power. Today's governments are forced to contend with their difficult position as hosts and cheerleaders for private corporations—lending their military power, diplomatic capital, and economic heft to promote their homegrown interest—and with their simultaneous role as supporters of the everyman who harbors suspicions of monopolistic greed and secrecy. That may prove an untenable contradiction. And, as detailed earlier in this book, the emergence of AI will make governance by any established institution more difficult.

Meanwhile, corporations could form alliances to consolidate their already considerable strength. Those alliances might be built on complementary advantages and the profit of amalgamation or, alternatively, on a shared philosophy of development and deployment of AI systems. These corporate alliances might take on traditional nation-state functions, though rather than seeking to define and expand bounded territories, they would cultivate diffuse digital networks as their domains.

And there is still another alternative. Uncontrolled, open-source diffusion could give rise to smaller gangs or tribes with substandard but substantial AI capacity, sufficient to administer to, provide for, and defend themselves within some limited scope. Among human groups that reject established authority in favor of decentralized

finance, communication, and governance, proto-anarchy could win out. Or such groupings might incorporate a religious dimension, perhaps driven by one of the conceptions of AI and divinity explored in an earlier chapter. After all, in terms of reach, Christianity, Islam, and Hinduism have all been larger and longer-lasting than any state in history. In the age to come, religious denomination, more than national citizenship, might conceivably prove the more relevant framework for identity and loyalty.

In either future, whether dominated by corporate alliances or diffused into loose religious groupings, the new "territory" that each group would claim—and over which they would fight—would not be inches of land but perhaps digital devices, as a signal of the loyalties of individual users. Linkages among these users and any administration—undoubtedly affected by the complex effect of AI on the status of traditional centralized government—would subvert the traditional notion of citizenship, and agreements among the entities would be unlike ordinary alliances.

Historically, alliances have been forged by individual leaders and have served to augment a nation's strength in case of war. By contrast, the prospect of citizenships and alliances—and perhaps conquests or crusades—structured around the opinions, beliefs, and subjective identities of ordinary people in times of peace would require a new (or very old) conception of empire. It would also force a reassessment of the obligations entailed in pledging allegiance and the cost of exit options, if indeed any were to exist in our AI-entangled future.

PEACE AND POWER

The foreign policies of nation-states, and thus international systems, have been built and then adjusted by balancing idealism and realism. The temporary balances struck by our leaders are seen in retrospect not as end states but as only ephemeral (if necessary) strategies for their time. With each new age, this tension has

produced a different expression of what constitutes political order. A leader cannot merely realize an option that falls along an existing and already-considered spectrum. Instead, governors must make at least some choices that derive (or appear to derive) from inspiration—frequently encouraging the pursuit of objectives that lie beyond the reach of practical attainment.

The dichotomy between the pursuit of interests and the pursuit of values—or between a particular nation-state's advantage and the global good—has been part of this unending evolution. In the conduct of their diplomacy, leaders of smaller states historically have responded straightforwardly, prioritizing the necessities of their own survival. By contrast, those responsible for global empires, with the means to realize additional goals, have faced a more agonizing predicament.

Since the beginning of civilization, as human units of organization have grown, they have simultaneously achieved new levels of cooperation. But today, perhaps due to the scale of our planetary challenges as well as to the material inequalities evident among and within states, a backlash against this trend has surfaced. Might AIs prove themselves commensurate with this still-grander scale of human governance, capable of seeing with granularity and fidelity the interplay of the globe and not merely the imperatives of the nation? Could they be relied upon to calculate—more precisely than we have ever done before—first our interests and our values, and then their correct proportion and relation to each other?

It would be unrealistic to expect, as one of us has put it before, that human leaders will reliably "confine our actions to situations in which our moral, legal, and military positions are completely in harmony and where legitimacy is most in accord with the requirements of survival."[17] For humans, this is still true. Yet we harbor a hope that AIs, deployed for political ends at home and abroad, might do more than just illuminate balanced trade-offs. Ideally, they could provide new, globally optimal solutions, acting on a longer time horizon and with higher resolution than humans are capable of, and thus bringing each of our competing human interests

into alignment. In the coming world, machine intelligences navigating conflict and negotiating peace might help clarify, or even surmount, our traditional dilemmas.

However, if AI were indeed to fix problems that we should have hoped to solve ourselves, we could face a crisis of confidence: that is, overconfidence on the part of some and lack of confidence on the part of others. To the former, once we understand the limits of our own ability for self-correction, it may be difficult to admit that we have come to cede too much power to the assumed wisdom of machines in handling existential issues of human conduct. To the latter, the realization that simply removing human agency from the handling of our affairs has been enough to solve our most intractable problems might reveal too explicitly the shortcomings of human design. If peace has always been but a simple voluntary choice, the price of human imperfection has been paid in the coin of perpetual war. To know that a solution has always existed but has never been conceived by us would be crushing to human pride.

This is one especially poignant instance of the dilemma of dependence—and subsequent perceived inferiority—explored in an earlier chapter. But, in the case of our security, unlike that of our displacement in scientific or other academic endeavors, we may more readily accept the impartiality of a mechanical third party as necessarily superior to the self-interestedness of a human—just as humans easily recognize the need for a mediator in a contentious divorce. It is our belief, and hope, that in this case some of our worst traits will enable us to exhibit some of our best: that the human instinct toward self-interest, including at the expense of others, may prepare us for accepting AI's transcendence of the same.